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Project Title:	Elucidating the Relationship Between the Effects of Various Radiation Qualities and Cancer Development Processes using Novel Flow-based Assays		
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Zip Code:	89154-9900	**Congressional District:**	1
Comments:	NOTE: Formerly at Lawrence Berkeley National Laboratory until fall 2017.		
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No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
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Flight Program:			
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Task Description:	Current radiation quality factors used for determining cancer risk for space radiation do not provide the accuracy needed, and there is debate as to the relevance of the experimental studies they're based upon. Therefore, one of NASA's primary goals is to improve the understanding of the carcinogenic process related to high LET exposures to better define the risk. Recently our current understanding of carcinogenesis has improved, and a number of cancer hallmarks have been identified. Some hallmarks, or surrogate cancer endpoints include genetic instability, telomere length changes and aberrant cellular signaling. Identifying relationships between early and late cell signaling profiles, later surrogate cancer endpoints, and radiation qualities will aid in determining a better risk assessment for cancer formation. In addition, using relevant cell types, such as human epithelial cells in which the majority of cancers develop is also key. In this proposal we will study critical phosphorylation signaling at early and late time points as well as surrogate cancer endpoints of		

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telomere length changes and genomic stability as measured by centrosome amplification, at these same late times post exposure. Epithelial and fibroblast cells from the same individuals will be used to better understand the etiology of cancer, and why certain cell types more easily become carcinogenic. We will also perform studies to understand the heterogeneous nature of high LET radiation. Energy deposition from high LET ions is dense in the track cores, resulting in closely spaced and complex damages, and delta rays originating from cores provide additional exposures similar to X-rays. Using carefully chosen charge, energy and LETs of particles to obtain differing populations of cells hit by tracks and delta rays alone we will investigate how phosphorylation profiles change. These studies will allow us to better quantify the effects of space radiation and its relationship to the carcinogenic process.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

Introduction:
In this proposal we are studying critical phosphorylation signaling at early (< 24 h) and late time points (days-weeks) following exposure. In addition, we will determine the relationship between these phospho-protein patterns, and two indicators of genomic stability and cancer progression, telomere shortening and centrosome amplification. Breast cancer studies have revealed that telomere length decreases during progression from usual ductal hyperplasia (UDH) to invasive cancer, thus telomere length changes can provide a readout of the carcinogenic process. A second marker we will be following after exposure is centrosome amplification which involves gains and losses of whole chromosomes and is believed to be due to continuous chromosome mis-segregation during mitosis resulting from centrosome defects. Combinations of increases in number, size, abnormal structure and function characterize centrosome amplification. Centrosome amplification has been detected in a broad range of solid tumors and more recently in leukemia and lymphoma. Centrosome abnormalities have been shown to increase with increasing tumor grade and aggressiveness. Over-expression of pericentrin, a centrosome protein, mimics many of the cellular characteristics of a high-grade carcinoma. Therefore centrosome abnormalities are also a very good indicator of cancer progression. Both of these potential markers for carcinogenesis will be followed using flow based assays.

A difficulty with high-LET exposures, particularly at low doses, is that very uneven levels of damage are deposited within the population of cells. When studying effects at longer times post low dose exposure healthy cells will outgrow the population making it difficult to study effects in the minor population more severely affected. To study populations of cells exhibiting varying levels of damage we will isolate and individually expand populations separately for analysis of surrogate carcinogenic endpoints. We will utilize a reagent called CFSE which allows one to identify cells blocked at various cell cycles following damage, and thus isolate more heavily damaged cells based on cell cycle delays. Sets of mammary epithelial and fibroblast cells from the same individuals will be used to better understand the etiology of cancer, and whether certain cell types more frequently show evidence of genetic instability upon radiation exposure. These studies will complement our earlier studies using a normal hTERT immortalized skin fibroblast line, and help us identify unique signaling patterns specific to different cell types. By using two cell types from the same individual we will identify genotype influences in response. The relevance of considering d-rays when determining the risk of high-LET radiation will also be studied. Carefully chosen particles of various energies will be used to obtain differing populations of cells hit in a greater proportion by tracks or d-rays to investigate how phosphorylation profiles and carcinogenic endpoints are affected by these components. The intent of this proposal is to provide a comprehensive picture of HZE effects from initial phosphorylation patterns to later phosphorylation profiles, and the relationships to various surrogate markers for carcinogenesis. Studies will be performed using novel flow cytometry-based assays, which have been optimized in our lab, to more rapidly and less subjectively monitor these endpoints.

Materials & Methods:

Cells were set up in flasks or chamber slides prior to irradiation, irradiated at the various doses and then fixed at multiple time points thereafter. A methanol and a combination paraformaldehyde and methanol fixation were used. Fixed slides were sent back to our home institution in PBS to stain with appropriate primary and secondary antibodies. Cells in flasks were trypsinized off and fixed, then sent back on ice in methanol. Aliquots of fixed cells are then stained for various phospho-proteins and analyzed via flow cytometry.

Results/ Discussion: During our first year of funding we have completed two BNL runs and collected data at each run for two cell types, normal fibroblasts (82-6) and mammary epithelial cells (184 diamond). Data has been collected following exposure to Fe at 300 MeV/u, Fe at 1000 MeV/u, and Si at 93 MeV/u. We have completed flow cytometry staining and analysis for 82-6 normal fibroblasts for all these ions and energies with samples from the first run, and are currently in the process of staining and analyzing samples from the second run. Phospho-kinetics for gamma H2AX, a marker of DSBs, pATF2, a DNA damage response protein, and pSMC1 another DNA damage response protein have been obtained for early timepoints (0.5, 2, 4, 8 and 24 h) following multiple doses (0.05, 0.1, 0.5, and 2 Gy). Our previous studies indicated that exposure to Fe ion (1000 MeV/u) resulted in high levels of gamma H2AX, yet low levels of pATF2 and pSMC1. Whereas, following Xray exposure we observed more equal levels of induction for each phospho-protein. These results suggested that low LET radiation provided a greater signal for ATM activation than did high LET exposure. Based on these results we expected that when we altered radiation quality such that less delta rays were received by the cell we may then see a reduction in the levels of ATM-mediated phosphorylations as compared to what we had seen with Fe @ 1000 MeV/u (~ 50% of its dose is thought to result from delta ray components and 50% from the direct track high LET core). However unexpectedly all high LET radiations induced similar levels of pATF2 and pSMC1 at 0.5 h post exposure regardless of Z2/ß2 parameters for the radiation qualities. Differences between high LET exposures were noted for gamma H2AX, with Fe 1000 MeV/u being the most effective, then Fe 300 MeV/u, Xray and finally Si 93 MeV/u. Based on the track structure theory each charge Z would have a different value of LET where peak effectiveness occurs. Iron should be somewhere between 150 and 200 and Si somewhere between 100 and 150 keV/µ. So in this experiment Si is over the peak (LET = 150 keV/µ) and Iron near the peak (LET =150 KeV/µ) and 300 MeV/u Iron is even further over the peak (LET = 240 KeV/µ) and thus should be less effective than 1 GeV/u Iron. So DSB induction, or gamma H2AX levels, seem to generally follow this pattern with the exception that Si would have been expected to be more effective than Xrays and perhaps more effective than 300 MeV/u Fe. More unexpected were the results for Si at our latest “early” time point of 24 h. We observed a notable induction in both pATF2 and pSMC1 at 24 h which was greater than what was seen with the most effective particle for gamma H2AX induction, 1 GeV/u Iron. At 24 h post exposure based on remaining gamma H2AX levels nearly all DSBs are repaired, so it seems some other signaling event(s) unrelated to DSBs are causing this induction in ATM-mediated phosphorylations following Si 93 MeV/u exposure. This implies that these low energy ion exposures may be more damaging to the cell at later

	timepoints, or initiate events within the cell to a much greater extent than higher energy exposures. Thus these studies are important in providing critical information to aid in understanding the cellular effects of various radiation quality exposures, and will aid in assessing the risk for astronauts exposed to these particles. We have just completed a second independent experiment with all these ions and energies collecting samples at all of these timepoints/doses and are currently analyzing these results. Given that this was the first year of funding, some time has been needed for optimization of conditions for growing, FACS staining and analysis using the new breast epithelial cells. These cells normally have more auto-fluorescence than fibroblast cells so we have had to find ways to limit this in order to allow us to visualize subtle changes in phospho-protein levels via flow cytometry. We have recently optimized conditions for nuclear isolation of fixed samples so that we can eliminate non-specific fluorescence from the membrane and cytoplasm and will begin using this procedure for staining of fixed samples from the first two runs. We have also performed some preliminary telomere and centrosome FACS-based assay experiments and noted similar problems with breast epithelial cells and are using a similar stratedgy to eliminate non-specific fluorescence for these assays as well. Awards, publications, presentations: 1) Presented at NASA Human Research Program Investigators Workshop, Feb 2-4, 2009, League City, Texas. Talk called “Cell-type Specific and Radiation-quality Dependent Phospho-profile Responses.” Mary K. Whalen, Sukhleen K. Gurai, Hengameh Zahed-Kargaran and Janice M. Pluth. Future Research Plan: In the forthcoming year we plan to finish the analysis of slides and FACS samples for 82-6 normal fibroblasts and 184 diamond mammary epithelial cells, staining for phospho-protein antibodies noted above. Results will be averaged with a third independent experiment completed in the upcoming year. We will perform experiments to determine telomere length changes and frequencies of centrosome defects in normal fibroblasts and breast epithelial cells. We also plan to perform some preliminary optimization experiments following high LET exposures using the CFSE component. Cellular addition of CFSE to growing cultures will allow allow us to separate cells based on cell cycle delay on various days post exposure, and provide a very accurate assesment of how various radiation qualites affect cell cycle.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Abstracts for Journals and Proceedings	Whalen MK, Gurai SK, Zahed-Kargaran H, Pluth JM. "Cell-type Specific and Radiation-quality Dependent Phospho-profile Responses." Presented at NASA Human Research Program Investigators' Workshop, League City, Texas, Feb 2-4, 2009. NASA Human Research Program Investigators' Workshop, League City, Texas, Feb 2-4, 2009. , Feb-2009